TuneMyBG LLM reproducibility — plain-English summary

Package generated 2026-08-21 covering 2026-08-07 to 2026-08-21 (14 days). Glucose: mean 121.6 mg/dL, time in range 86.0%, below range 5.1%, CV 32.6%. Carb entries logged: 0.

Each run is a brand-new conversation: package attached with Prompt 1, then Prompts 2, 3 and 4 in turn, then the app's correction prompt once if the JSON failed the app's checks. AAPS defaults are from nightscout/AndroidAPS master @ 598e2eb (2026-08-02).

claude-haiku-4-5 — 27 runs (`real_haiku45_50`)

At a glance

- App outcome over 28 completed runs: accepted first time 13; accepted after the repair prompt 0; invalid JSON with no prompt offered 2; schema stop with no prompt offered 0; still rejected after repair 13. Dead ends (no way forward in the app): 2.
- Strict compliance with Prompt 4's own rules (which the app does not fully check): 1/27 (4%) of runs first time.
- Settings changed per run: 0–5 of 23 (most runs: 1).
- 23 settings were left unchanged in every single run; 13 were changed in at least one run, of which 0 were changed in more than half the runs.
- Headline profile recommendation focus: basal in 12/27 (44%), ISF in 1/27 (4%), basal_cr_isf_dia_validation_sequence in 1/27 (4%), basal, target, CR, ISF, DIA in 1/27 (4%), carb logging, then isf verification in 1/27 (4%), carbohydrate_ratio_and_meal_strategy in 1/27 (4%), insulin_duration in 1/27 (4%), carbohydrate_logging_and_meal_bolus_execution in 1/27 (4%), DIA and basal timing (safety priority), conditional midday basal increase (medium priority, contingent on ISF verification) in 1/27 (4%), basal_and_targets_and_cr_and_isf_and_dia in 1/27 (4%), carb_ratio in 1/27 (4%), isf in 1/27 (4%), meal_strategy_and_carb_logging in 1/27 (4%), basal, cr, isf, dia, target in 1/27 (4%), meal_bolus_timing_and_carbohydrate_entry in 1/27 (4%), Profile defects: flat ISF 56 mg/dL/U (too high, masked by Dynamic ISF compression to 2–4); overnight basal 0.6 U/h (excessive, causing hour 1 low 90.6 mg/dL, 18.5% below range); afternoon CR 13.2 g/U (loosest segment, coinciding with sustained hyperglycemia 130–162 mg/dL, 26% above range). GLP-1 post-2026-08-14 extends meal absorption to 240–360m, but CR and ISF unchanged; reactive bolusing (no prebolus, no carb logging) allows meal boluses to peak at 60m while carbs continue absorbing, causing rebounds to 188–235 mg/dL. AAPS automation (SMB, UAM, temp basal, Dynamic ISF, safety limits) is appropriately configured; no automation defects. Automation is compensating for profile-level errors in real-time. in 1/27 (4%).
- Profile recommendation decision: change ×17, keep ×10.

Changed in at least one run

Frequency is how many of the runs suggested a different value for that setting. Range is the spread of the suggested values across those runs.

Setting	Current	AAPS default	Suggested change in	Suggested values	Direction
Profile / basal 10:00	0.7 U/h	—	10/27 (37%)	0.8–0.85 U/h; most often 0.8 U/h (6 of 10); 2 distinct values	higher
Profile / basal 16:00	0.5 U/h	—	6/27 (22%)	0.6–0.7 U/h; most often 0.7 U/h (3 of 6); 3 distinct values	higher
Profile / basal 00:00	0.6 U/h	—	3/27 (11%)	0.5–0.55 U/h; most often 0.5 U/h (2 of 3); 2 distinct values	lower
Profile / DIA	10 h	— (AAPS defaultDIA = 5.0 h (Constants.kt))	3/27 (11%)	5.5–6 h; most often 5.5 h (2 of 3); 2 distinct values	lower
Profile / ISF 00:00	56 mg/dL/U	—	3/27 (11%)	40–110 mg/dL/U; most often 90 mg/dL/U (1 of 3); 3 distinct values	mixed
AAPS / carbohydrate absorption model / min_5m_carbimpact	8 mg/dL/5 min	8 mg/dL/5 min	2/27 (7%)	5–12 mg/dL/5 min; most often 12 mg/dL/5 min (1 of 2); 2 distinct values	mixed
AAPS / max basal and max IOB limits / max_iob_units	25 U	3.0 U (Simple Mode recalculates this from the profile)	2/27 (7%)	8–20 U; most often 8 U (1 of 2); 2 distinct values	lower

Setting	Current	AAPS default	Suggested change in	Suggested values	Direction
AAPS / Autosens and Dynamic ISF / dynamic_isf_adjustment_factor_percent	70 %	100 %	1/27 (4%)	always 85 %	higher
Profile / basal 19:00	0.7 U/h	—	1/27 (4%)	always 0.6 U/h	lower
Profile / CR 08:00	13.2 g/U	—	1/27 (4%)	always 11.5 g/U	lower
Profile / CR 16:00	10.5 g/U	—	1/27 (4%)	always 9.5 g/U	lower
Profile / target 00:00	90 mg/dL	—	1/27 (4%)	always 100 mg/dL	higher
Profile / target 23:00	90 mg/dL	—	1/27 (4%)	always 100 mg/dL	higher

Kept unchanged in every run

- **Basal rates**: 05:00 0.7 U/h; 06:00 0.8 U/h; 18:00 0.6 U/h
- **Carb ratios**: 00:00 13 g/U; 04:00 11.1 g/U; 20:00 12.6 g/U
- **Targets**: 08:00 100 mg/dL
- **AAPS smb_uam**: smb_enabled = true; uam_enabled = true
- **AAPS smb_activation**: smb_always_enabled = true; smb_with_cob_enabled = false; smb_after_carbs_enabled = false; smb_with_temp_target_enabled = true; smb_with_high_temp_target_enabled = false
- **AAPS smb_delivery**: smb_interval_minutes = 3 min; max_smb_basal_minutes = 15 min; max_uam_smb_basal_minutes = 20 min
- **AAPS safety_limits**: max_basal_u_per_hour = 12 U/h
- **AAPS sensitivity**: autosens_enabled = false; dynamic_sensitivity_enabled = true; autosens_raises_target = false; autosens_lowers_target = false
- **AAPS carb_absorption**: meal_max_absorption_hours = 7 h

Primary recommendation (first item the model listed)

By area: basal ×9, meal_strategy ×4, ISF ×1, Data and meal strategy ×1, safety ×1, meal strategy ×1, Carbohydrate logging (highest-leverage behavior change) ×1, meal_logging ×1, dia ×1, behavioral_and_logging ×1, Data Integrity – ISF Verification ×1, carb_logging ×1, isf ×1, meal_bolus_strategy ×1, meal_carbs ×1, profile ×1.

- Correct ISF underestimation from 56 to 90 mg/dL/U ×1
- Restart carb logging immediately for 3 complete days ×1
- Reduce max_iob from 25 U to 6-8 U immediately ×1
- Enable carb logging (Nightscout treatment entries) immediately for all meals and snacks; log carb amount (grams) and meal time. Continue for minimum 7 days to establish baseline. ×1
- Reduce basal 00:00 from 0.6 U/h to 0.55 U/h or 0.5 U/h to address hour 1 lows (18.5% TBR, nadir 42–55 mg/dL). ×1
- Raise basal in 10:00–16:00 afternoon window to address persistent hyperglycemia. ×1
- Implement meal carbohydrate logging ×1
- Reduce overnight and evening basal segments to address recurrent hypoglycemia ×1
- … and 19 other titles